

SECTION 7 – BIG-O ANALYSIS (Part VIII)

Smart Campus Student Resource Management System – NTC CC105 Data Structure

Explain, in your own words, why each stated complexity applies to the implementation.

Operation	Time Complexity	Explain Why
Direct array access	$O(1)$	A specific array element is accessed directly using its index. The number of elements does not change the access time.
Array traversal	$O(n)$	The program visits each element once. With n elements, the loop performs about n iterations.
Linear search	$O(n)$	The program checks elements one by one until the target is found or all elements have been checked. In the worst case, all n elements are examined.
Matrix traversal ($n \times n$)	$O(n^2)$	Nested loops visit every row and every column. With n rows and n columns, there are $n \times n$ element visits.